

MATH9102

COMPARING NOMINAL VARIABLES

Sources used in creation of this lecture:

Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using IBM SPSS, Andy Field; Understanding Basic Statistics, Brase and Brase; SPSS Survival Manual, Julie Pallant

Comparing Nominal Variables

When two nominal variables are independent, there is no relationship between them.

This means that

- For all cases
 - The classification of a case into a particular category of one variable (the group variable) has no effect on the probability that the case will fall into any particular category of the second variable (the test variable).

Comparing Nominal Variables

Analyzing categorical variables

- Central Tendency (mean/median) of a categorical variable is meaningless
 - The numeric values you attach to different categories are arbitrary
 - The mean of those numeric values will depend on how many members each category has.
- Therefore, we analyze frequencies.

Comparing Nominal Variables

Using the **bullying.dat** dataset (Paul Connolly)

- Dataset Descriptor:
- <https://www.routledge.com/Quantitative-Data-Analysis-in-Education-A-Critical-Introduction-Using-SPSS/Connolly/p/book/9780415372985>
- Contains the results of national school level survey, subset of this survey which deals with bullying.

Question:

- “Is there a relationship between being bullied in school and participating in bullying of others in school?”
- H0: There is no relationship between being a victim of bullying in school and participating in bullying others in school.
- HA: There is a relationship between being a victim of bullying in school and participating in bullying others in school.
- Variables:
- ‘ubullsch’ – Q26. Have you yourself ever been bullied in school? {1- Yes, 2 –No, 99-not answered}
- ‘ubulloth’ - Q28 - Have you yourself ever taken part in bullying other students? {1- Yes, 2 –No, 99-not answered}

Comparing Nominal Variables

We could create a contingency table/

A → Students who were bullied and bullied others

B → Students who were bullied but did not bully others

C → Students who were not bullied but bullied others

D → Students who were not bullied and did not bully others

N → Total number of students

A Chi square test will compare the observed frequencies (A, B, C, D) with the expected frequencies under the assumption being bullied and bullying others are independent.

The chi-square statistic will be calculated, and its significance will be determined based on the p-value and the degrees of freedom (df = 1 in this case).

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A	B	A + B
Was Not Bullied	C	D	C + D
Total	A + C	B + D	N

Chi-square Test

We are comparing observed values to expected values according to a specific hypothesis (null hypothesis)

The null hypothesis for this test states that the proportions (the distribution across categories) are the same for all of the populations

Views the data as two (or more) separate samples representing the different populations being compared

- The same variable is measured for each sample by classifying individual subjects into categories of the variable.
- The data are presented in a matrix with the different samples defining the rows and the categories of the variable defining the columns.
- The data, called **observed frequencies**, simply show how many individuals from the sample are in each cell of the matrix.

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A	B	A + B
Was Not Bullied	C	D	C + D
Total	A + C	B + D	N

Chi Square Test for Independence

We are comparing observed values to **expected values** according to a specific hypothesis (null hypothesis)

- The null hypothesis for this test states that the proportions (the distribution across categories) are the same for all of the populations
- The null hypothesis is used to construct an idealized sample distribution of **expected frequencies** that describes how the sample would look if the data were in perfect agreement with the null hypothesis
- Calculated from the observed frequencies:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

Expected Frequencies

Suppose we had the following observed frequencies:

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A = 35	B = 209	A + B = 244
Not Bullied	C = 25	D = 530	C + D = 555
Total	A + C = 60	B + D = 739	N = 799

	Formula	Expected Frequency
E(A) = “Was Bullied” & “Bullied Others”	$(244 \times 60) / 799$	18.32
E(B) = “Was Bullied” & “Did Not Bully”	$(244 \times 739) / 799$	225.68
E(C) = “Not Bullied” & “Bullied Others”	$(555 \times 60) / 799$	41.68
E(D) = “Not Bullied” & “Did Not Bully”	$(555 \times 739) / 799$	513.32

Comparing

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A = 35	B = 209	A + B = 244
Not Bullied	C = 25	D = 530	C + D = 555
Total	A + C = 60	B + D = 739	N = 799

Comparing these to the observed counts (35 vs. 18.32, 25 vs. 41.68, etc.) shows a clear deviation — especially:

- More students than expected were both **bullied and bullied others (A)**.
- Fewer students than expected were **not bullied but bullied others (C)**.

This is what our chi square test will measure and will test to see if the difference is large enough to be statistically significant.

	Formula	Expected Frequency
E(A) = “Was Bullied” & “Bullied Others”	$(244 \times 60) / 799$	18.32
E(B) = “Was Bullied” & “Did Not Bully”	$(244 \times 739) / 799$	225.68
E(C) = “Not Bullied” & “Bullied Others”	$(555 \times 60) / 799$	41.68
E(D) = “Not Bullied” & “Did Not Bully”	$(555 \times 739) / 799$	513.32

Chi-Square Test for Independence

A chi-square statistic is computed to measure the amount of discrepancy between the ideal sample (expected frequencies from H_0) and the actual sample data (the observed frequencies).

The output is a cross-tabulation table

In R

```
#CrossTable(predictor, outcome, fisher = TRUE, chisq = TRUE, expected =
TRUE)#Note: for this approach there must be the same number of values for
both variables so#missing data is important.

# Remove rows where either variable is NA

bully_clean <- bully %>%  filter(!is.na(ubullsch) &
!is.na(ubulloth))

# Now run CrossTable requesting both Fisher and Chi Square so we have full
information# We request the format in SPSS to make it easier to interpret.

gmodels::CrossTable(
bully_clean$ubullsch,
bully_clean$ubulloth,
fisher = TRUE,
chisq = TRUE,
expected = TRUE,
sresid = TRUE,
format = "SPSS")
```

Count
Expected Values
Chi-square contribution
Row Percent
Column Percent
Total Percent
Std Residual

bully_clean\$bullysch	bully_clean\$bullyoth	Row Total
No	Yes	
No	530	25
	513.323	41.677
	0.542	6.673
	95.495%	4.505%
	71.719%	41.667%
	66.333%	3.129%
	0.736	-2.583
Yes	209	35
	225.677	18.323
	1.232	15.179
	85.656%	14.344%
	28.281%	58.333%
	26.158%	4.380%
	-1.110	3.896
Column Total	739	60
	92.491%	7.509%

Make sure your %s add up to 100.

Pearson's Chi-squared test

Chi² = 23.62668 d.f. = 1 p = 1.169546e-06

Pearson's Chi-squared test with Yates' continuity correction

Chi² = 22.2312 d.f. = 1 p = 2.417134e-06

Fisher's Exact Test for Count Data

Sample estimate odds ratio: 3.543753

Alternative hypothesis: true odds ratio is not equal to 1

p = 3.65894e-06

95% confidence interval: 2.006207 6.341761

Alternative hypothesis: true odds ratio is less than 1

p = 0.9999992

95% confidence interval: 0 5.799364

Alternative hypothesis: true odds ratio is greater than 1

p = 3.154022e-06

95% confidence interval: 2.184139 Inf

Minimum expected frequency: 18.3229

Need to check this – assumption is that lowest expected frequency is 5 or more. If not then this test is not appropriate and should use Fisher's Exact Probability Test.

If we are looking at 2x 2
table we look at Yates
continuity correction

Pearson's Chi-squared test

chi^2 = 23.62668 d.f. = 1 p = 1.169546e-06

Pearson's Chi-squared test with Yates' continuity correction

chi^2 = 22.2312 d.f. = 1 p = 2.417134e-06

Fisher's Exact Test for Count Data

Sample estimate odds ratio: 3.543753

Alternative hypothesis: true odds ratio is not equal to 1
p = 3.65894e-06
95% confidence interval: 2.006207 6.341761

Alternative hypothesis: true odds ratio is less than 1
p = 0.9999992
95% confidence interval: 0 5.799364

Alternative hypothesis: true odds ratio is greater than 1
p = 3.154022e-06
95% confidence interval: 2.184139 Inf

Yate's Continuity Correction

This is to help prevent us making a Type I error

When we are doing a Chi-square test, we are assuming that the probability distribution of the binomial frequencies observed approximate the Chi-square distribution

- This is not quite correct
- Yate's correction adjusts Pearson's Chi-square by subtracting 0.5 from each of the differences between the observed values and the expected values, which will reduce the Chi-square statistic and its associated p-value
- Particularly important for small data (larger datasets this can be overcome)
- May tend to overcorrect – make sure you reflect your research area's perspective (some would say Yates is unnecessary even for small data).

In R – more simplistic

```
#Create your contingency table
```

```
mytable<-xtabs(~ubullsch+rsex, data=bully)
```

```
ctest<-chisq.test(mytable, correct=TRUE)#chi square test
```

```
#correct=TRUE to get Yates correction needed for 2x2 table
```

```
ctest#will give you the details of the test statistic and p-value
```

```
ctest$expected#expected frequencies
```

```
ctest$observed#observed frequencies
```

```
ctest$p.value
```

```
OUTPUT:
```

```
X-squared = 2.3005, df = 1, p-value = 0.1293
```

Chi Square Statistic

Measures the amount of discrepancy between the ideal sample (expected frequencies from H_0) and the actual sample data (the observed frequencies = f_o).

- A large discrepancy results in a large value for chi-square which indicates that the data do not fit the null hypothesis and the hypothesis does not hold.
- We compare our test statistic to the Chi Square distribution (in the form of a table) and we can get the probability that any deviation in the observed from the expected is due to chance only

For one degree of freedom

- The critical value associated with $p = 0.05$ for Chi Square is 3.84
- The critical value associated with $p = 0.01$ it is 6.64
- *Chi-square values higher than this critical value are associated with a statistically low probability that H_0 holds.*

Effect Sizes

Phi (2 x 2 contingency table)

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

0.1 is considered a small effect

0.3 a medium effect

0.5 a large effect.

Phi is used for 2x2 tables

Effect Sizes

Cramers V depends on the degrees of freedom

Cramer's V is suitable for tables larger than 2x2, and

$$V = \sqrt{\frac{\chi^2}{n \cdot df^*}}$$

<i>df</i>	<i>small</i>	<i>medium</i>	<i>large</i>
1	.10	.30	.50
2	.07	.21	.35
3	.06	.17	.29
4	.05	.15	.25
5	.04	.13	.22

Chi-Square Test for Independence

Reporting the findings

A Chi-square test of independence was conducted to examine the relationship between being bullied in school and bullying others. The association was statistically significant ($\chi^2(1, N = 799) = 22.23, p < .001$). Students who reported being bullied were significantly more likely to report bullying others (14.3%) than those who were not bullied (4.5%). The effect size, measured by Phi ($\phi = 0.17$), indicates a small association. There is therefore evidence to support rejecting the null hypothesis in favour of the alternate hypothesis.

Where did the %s come from?

bully_clean\$bullysch	bully_clean\$bullyoth		Row Total
	No	Yes	
No	530	25	555
	513.323	41.677	
	0.542	6.673	
	95.495%	4.505%	69.462%
	71.719%	41.667%	
	66.333%	3.129%	
	0.736	-2.583	
Yes	209	35	244
	225.677	18.323	
	1.232	15.179	
	85.656%	14.344%	30.538%
	28.281%	58.333%	
	26.158%	4.380%	
	-1.110	3.896	
Column Total	739	60	799
	92.491%	7.509%	

Bullied students who bullied others:

- $(35/244) \times 100 = 14.3\%$

Bullied students who did not bully others:

- $(209/244) \times 100 = 85.7\%$

Where did the %s come from?

bully_clean\$bullysch	bully_clean\$bullyoth		Row Total
	No	Yes	
No	530 513.323 0.542 95.495% 71.719% 66.333% 0.736	25 41.677 6.673 4.505% 41.667% 3.129% -2.583	555 69.462%
Yes	209 225.677 1.232 85.656% 28.281% 26.158% -1.110	35 18.323 15.179 14.344% 58.333% 4.380% 3.896	244 30.538%
Column Total	739 92.491%	60 7.509%	799

Non-Bullied students who bullied others:

- $(25/555) \times 100 = 4.5\%$

Non-Bullied students who did not bully others:

- $(530/555) \times 100 = 95.5\%$

Repeated Measures Categorical Variables

Use McNemar's test

Matched Samples or repeated measures
(pre-test/post-test)

- Two variables one recorded at Time 1 and one recorded at Time 2 (after an intervention)

Use for categorical variables with two
response options

- For 3 or more use Cochran's Q-test

In R: `mcnemar.test`